

STAT-2311

Analysis by Previous Question Chapter: 07(Suggestion)

Probability Distribution

1. Binomila Distribution
2. Poisson Distribution
3. Normal Distribution

What is Binomial Distribution?

- A discrete random variable X is said to have binomial distribution if its probability function is as follows:

$$f(x) = n_{C_x} p^x q^{n-x} ; n = 0, 1, 2, \dots \dots \dots x^n$$

Where n and p are the parameters of the distribution and $p + q = 1$

What is Poisson Distribution?

- A discrete random variable X is said to have Poisson distribution if its probability function is as follows:

$$f(x) = \frac{e^{-m} m^x}{x!} ; x = 0, 1, 2, \dots \dots \dots \infty$$

Where, m is the parameter and $e = 2.178$

What is Normal Distribution

- A continuous random variable X is said to have normal distribution if its probability density function is as follows:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}; \quad \begin{matrix} -\infty < x < \infty \\ -\infty < \mu < \infty \\ \sigma^2 \geq 0 \end{matrix}$$

Where μ and σ^2 are the parameters of the distribution

Important of normal distribution in computer science.

Ans: The normal distribution is vital in computer science because:

1. Data Analysis: Many datasets approximate normal distribution, aiding in predictive modeling.
2. Machine Learning: Used in algorithms like SVM and Gaussian Mixture Models.
3. Statistical Testing: Forms the basis for hypothesis tests and confidence intervals.
4. Simulations: Essential in generating realistic random data (e.g., Monte Carlo).
5. Signal & Image Processing: Models noise and applies Gaussian blur.

Conditions Under Which a Binomial Distribution Reduces to a Poisson Distribution

Ans: The binomial distribution reduces to the Poisson distribution under the following conditions:

1. Large number of trials (n): The number of trials n is very large.
2. Small probability of success (p): The probability of success p in each trial is very small.
3. Constant mean (λ): The product $\lambda = n \cdot p$ (mean of the binomial distribution) remains constant as $n \rightarrow \infty$ and $p \rightarrow 0$.

Conditions for Approximating a Binomial Distribution as a Normal Distribution

Ans: A binomial distribution can be approximated by a normal distribution when:

1. Large number of trials (n): The number of trials n is sufficiently large.

2. Moderate probability of success (p): The probability p is not too close to 0 or 1.
3. Satisfying the rule of thumb: $n \cdot p \geq 5$ and $n \cdot (1-p) \geq 5$.

Math Problem:

Problem:-1. A family has 6 children. The probability being a boy or a girl equal. Find the probability that the family has.

1. No girl
2. Excatly 2 girls
3. At most 2 girls
4. At least 4 girls

Solution:

Here $n = 6$

$P = q = \frac{1}{2}$ (The probability being a boy or a girl equal)

We know,

$$f(x) = n_{C_x} p^x q^{n-x} ; n = 0, 1, 2, \dots \dots \dots x^n$$

$$f(x) = 6_{C_x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{6-x} ; n = 0, 1, 2, \dots \dots \dots x^n$$

1. No girl

$$\begin{aligned} f(x=0) &= 6_{C_0} \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{6-0} \\ &= \frac{1}{64} \end{aligned}$$

2. Excatly 2 girls

$$\begin{aligned} f(x=2) &= 6_{C_2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{6-2} \\ &= \frac{15}{64} \end{aligned}$$

3. At most 2 girls

$$f(x \leq 2) = f(x = 0) + f(x = 1) + f(x = 2)$$

$$\begin{aligned} f(x = 1) &= {}^6C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{6-1} \\ &= \frac{3}{32} \end{aligned}$$

So,

$$\begin{aligned} f(x \leq 2) &= f(x = 0) + f(x = 1) + f(x = 2) \\ &= \frac{1}{64} + \frac{15}{64} + \frac{3}{32} \\ &= \frac{11}{32} \end{aligned}$$

4. At least 2 girls

$$f(x \geq 4) = f(x = 4) + f(x = 5) + f(x = 6)$$

$$\begin{aligned} f(x = 4) &= {}^6C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^{6-4} \\ &= \frac{15}{64} \end{aligned}$$

$$\begin{aligned} f(x = 5) &= {}^6C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^{6-5} \\ &= \frac{3}{32} \end{aligned}$$

$$\begin{aligned} f(x = 6) &= {}^6C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^{6-6} \\ &= \frac{1}{64} \end{aligned}$$

$$\begin{aligned}
 f(x \geq 4) &= f(x = 4) + f(x = 5) + f(x = 6) \\
 &= \frac{15}{64} + \frac{3}{32} + \frac{1}{64} \\
 &= \frac{11}{32}
 \end{aligned}$$

Problem:-2. If 2% of electric bulbs are manufactured by a certain company are defective. Find the probability that in a sample of 200 bulbs.

1. Exactly 2 bulbs
2. Less than 2 bulbs
3. More than 3 bulbs are defective

Solution:

We know,

The Probability function of a poisson distribution is,

$$f(x) = \frac{e^{-m} m^x}{x!} ; x = 0, 1, 2, \dots \dots \dots \infty$$

Now,

$$\begin{aligned}
 m &= np \\
 &= 200 \times \frac{11}{32} \\
 &= 4
 \end{aligned}$$

$$\text{Therefore, } f(x) = \frac{e^{-4} 4^x}{x!} ; x = 0, 1, 2, \dots \dots \dots \infty$$

1. Exactly 2 bulbs

$$\begin{aligned}
 f(x = 2) &= \frac{e^{-4} 4^2}{2!} \\
 &= 0.1464
 \end{aligned}$$

2. Less than 2 bulbs

$$f(x < 2) = f(x = 0) + f(x = 1)$$

$$\begin{aligned} f(x = 0) &= \frac{e^{-0} 4^0}{0!} \\ &= 0.0183 \end{aligned}$$

$$\begin{aligned} f(x = 1) &= \frac{e^{-1} 4^1}{1!} \\ &= 0.0732 \end{aligned}$$

$$\begin{aligned} f(x < 2) &= f(x = 0) + f(x = 1) \\ &= 0.0183 + 0.0732 \\ &= 0.0915 \end{aligned}$$

3. More than 3 bulbs are defective

$$\begin{aligned} f(x > 3) &= f(x = 0) + f(x = 1) + f(x = 2) + f(x = 3) + f(x = 4) + \\ &\quad f(x = 5) + \cdots + f(x = 200) \\ &= 1 - f(x < 4) \end{aligned}$$

$$\begin{aligned} f(x = 3) &= \frac{e^{-3} 4^3}{3!} \\ &= 0.1952 \end{aligned}$$

$$\begin{aligned} f(x < 4) &= 0.0183 + 0.0732 + 0.0915 + 0.1952 \\ &= 0.4331 \end{aligned}$$

$$f(x > 3) = 1 - f(x < 4)$$

$$= 1 - 0.4331$$

$$= 0.5669$$